Breast Ultrasound Image Classification using Transfer Learning and Class Imbalance Handling

DEEP LEARNING FOR MEDICAL IMAGING – CS328

ASSIGNMENT

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1. Introduction

Breast cancer is one of the most common cancers affecting women worldwide. Early detection significantly improves survival rates and treatment outcomes. Ultrasound imaging is widely used for detecting breast tumors because it is non-invasive, cost-effective, and safe. However, interpreting ultrasound images manually can be challenging and time-consuming.

Recent advancements in deep learning have enabled automated analysis of medical images. Convolutional Neural Networks (CNNs) have shown strong performance in image classification tasks. In this project, deep learning techniques are used to classify breast ultrasound images into three categories: Normal, Benign, and Malignant.

Two methodologies are implemented and compared:
1. Transfer Learning using MobileNetV2 (Baseline Model)
2. Transfer Learning with Class Imbalance Handling and Fine-Tuning

The objective is to evaluate the impact of class imbalance techniques on model performance while maintaining balanced precision and recall.

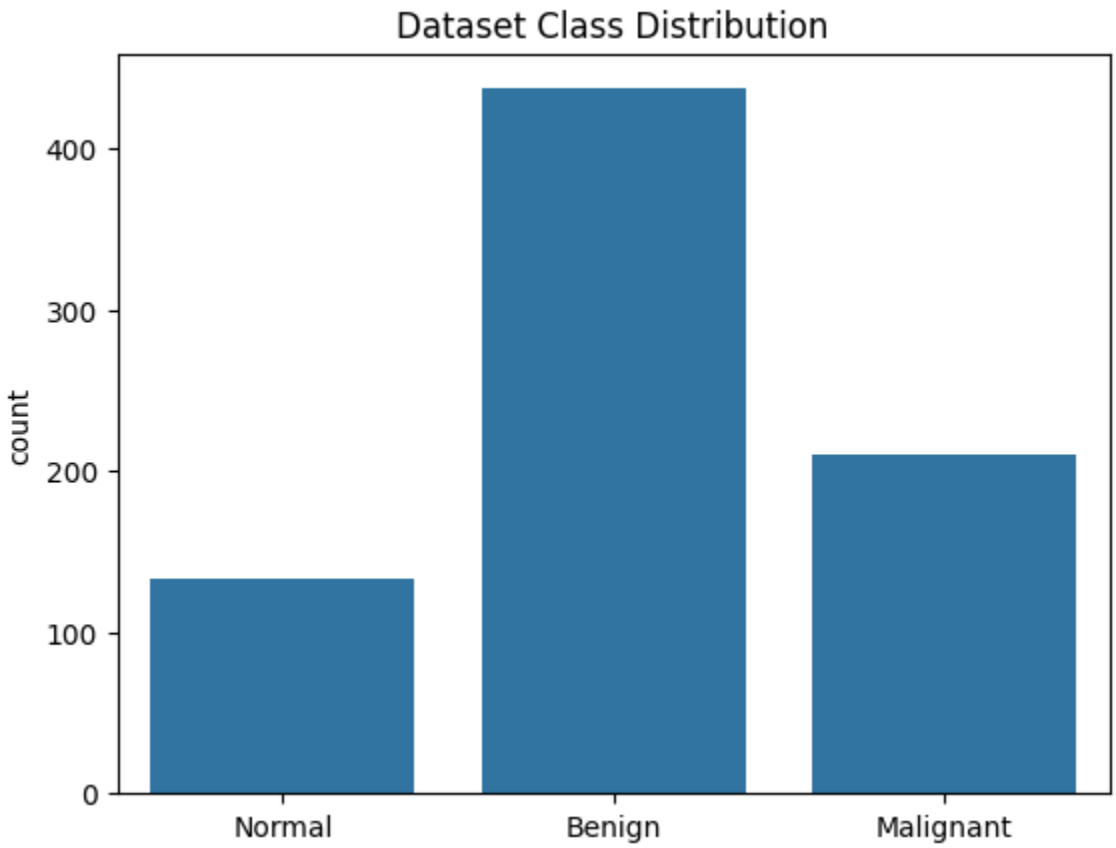
2. Dataset Description

The dataset used in this project is the BUSI (Breast Ultrasound Images) dataset. It contains ultrasound images categorized into three diagnostic classes.

Classes:
- Normal: No tumor present
- Benign: Non-cancerous tumor
- Malignant: Cancerous tumor

Dataset Size:
Normal: 133 images
Benign: 437 images
Malignant: 210 images
Total: 780 images

The dataset is class imbalanced, as benign samples are significantly more frequent than normal samples.



3. Data Preprocessing

Several preprocessing steps were performed before training the models:

• Images were resized to 224 × 224 pixels to match the input size required by MobileNetV2.
• Pixel values were normalized to the range [0,1] to stabilize neural network training.
• Mask images present in the BUSI dataset were removed to ensure only ultrasound images were used.
• The dataset was split using stratified sampling.

Dataset Split:
Training Images: 624
Testing Images: 156

4. Methodology 1 – Transfer Learning using MobileNetV2

The first methodology uses transfer learning with the MobileNetV2 architecture pretrained on the ImageNet dataset.

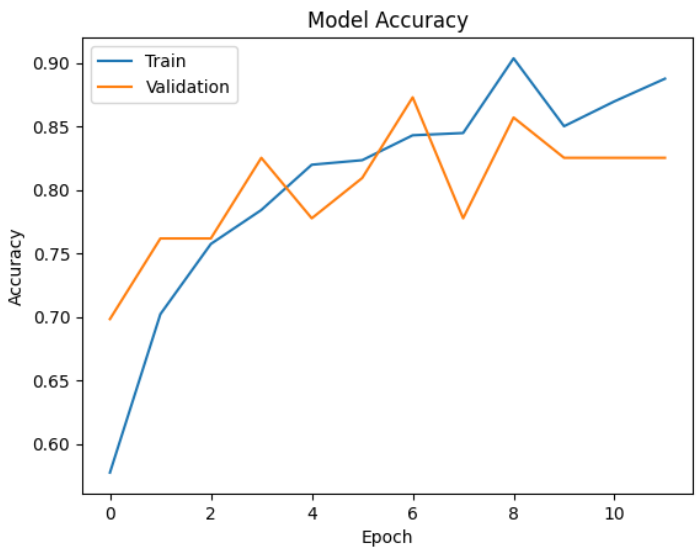
Transfer learning allows the model to use previously learned visual features such as edges, textures, and shapes.

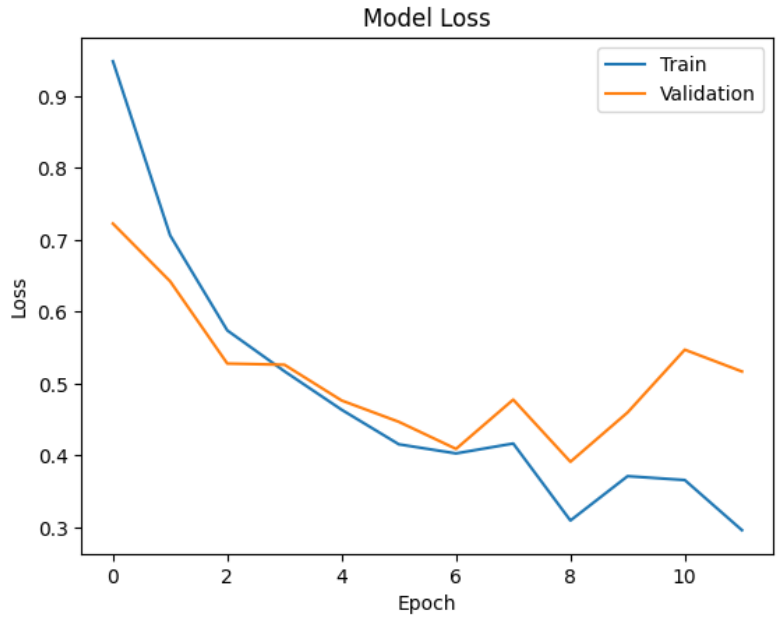
Model Architecture:
Input Image (224×224)
→ MobileNetV2 Feature Extractor (Frozen Layers)
→ GlobalAveragePooling2D
→ Dense Layer (128 neurons, ReLU activation)

→ Dropout Layer (0.5)
→ Softmax Output Layer (3 classes)

Training Parameters:
Optimizer: Adam
Loss Function: Categorical Crossentropy
Batch Size: 16
Epochs: 12

All layers of MobileNetV2 were frozen, and only the classification head was trained.





5. Methodology 2 – Transfer Learning with Class Imbalance Handling

The second methodology extends the baseline model by incorporating techniques to address dataset imbalance.

Techniques Used:

1. Data Augmentation
Random transformations such as rotation, zooming, horizontal flipping, and width/height shifting were applied to increase the variability of the training dataset.

2. Class Weighting
Class weights were used during training to give higher importance to minority classes.

Example class weights used:
Normal: 1.2
Benign: 1.0
Malignant: 1.3

3. Fine-Tuning MobileNetV2
Unlike the baseline model, the final layers of MobileNetV2 were unfrozen and fine-tuned so that the network could adapt to ultrasound-specific features.

Training Parameters:
Optimizer: Adam
Learning Rate: 0.00003
Epochs: 25
Early Stopping: Enabled

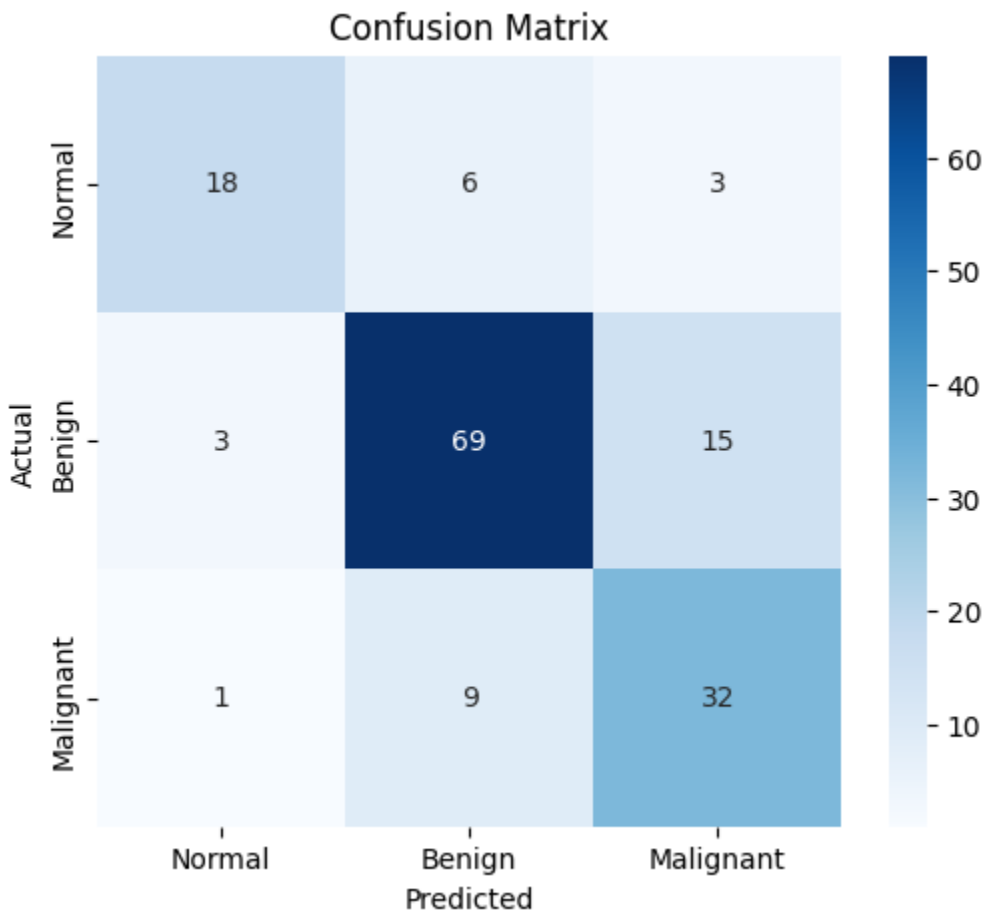
6. Results – Methodology 1

Test Accuracy: 0.76

Classification Report:

Normal → Precision: 0.82 Recall: 0.67 F1: 0.73
Benign → Precision: 0.82 Recall: 0.79 F1: 0.81
Malignant → Precision: 0.64 Recall: 0.76 F1: 0.70

Macro Average:
Precision: 0.76
Recall: 0.74
F1-score: 0.75



7. Results – Methodology 2

Test Accuracy: 0.72

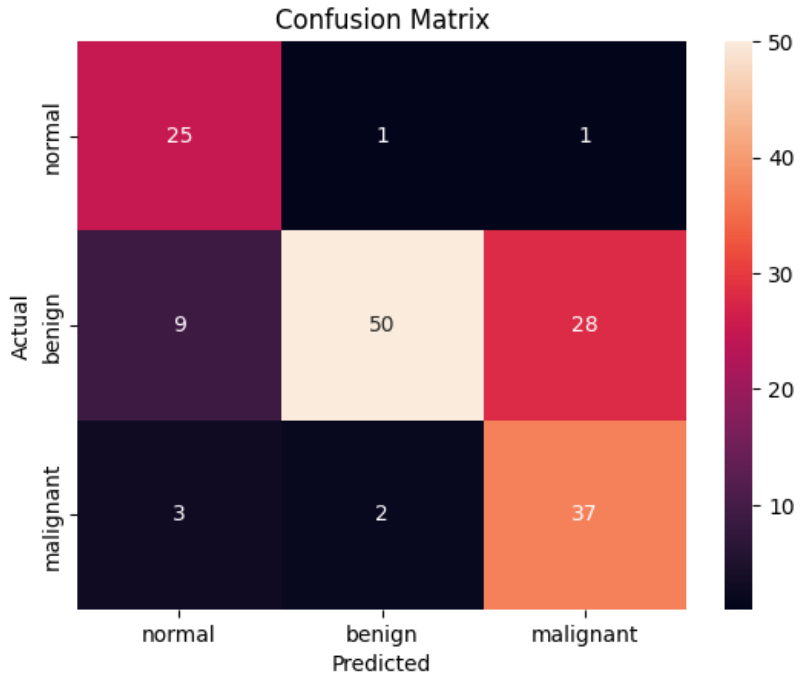
Classification Report:

Normal → Precision: 0.68 Recall: 0.93 F1: 0.78
Benign → Precision: 0.94 Recall: 0.57 F1: 0.71
Malignant → Precision: 0.56 Recall: 0.88 F1: 0.69

Macro Average:
Precision: 0.73

Recall: 0.79
F1-score: 0.73

The second methodology improves recall for malignant cases but slightly reduces overall accuracy.



8. Discussion

The baseline transfer learning model achieved higher overall accuracy, while the class imbalance handling approach improved recall for malignant tumors.

In medical diagnosis tasks, missing malignant cases can be critical. Therefore, improving recall for malignant samples can be beneficial even if overall accuracy decreases slightly.

The comparison between the two methodologies demonstrates how class imbalance handling can influence model predictions and improve detection of minority classes.

9. Conclusion

This project explored breast ultrasound image classification using deep learning techniques. Two methodologies were implemented: a baseline MobileNetV2

transfer learning model and an improved model incorporating class imbalance handling techniques.

The baseline model achieved higher overall accuracy, while the imbalance-aware model improved malignant tumor detection. The results highlight the importance of addressing dataset imbalance when developing medical image classification systems.

Future work could explore additional approaches such as focal loss, ensemble models, or larger datasets to further improve classification performance.